Edge-conditioned SEAL_directed | Detailed proposal

PROPOSAL · NOT IMPLEMENTED

A staged, leakage-safe route from relational edge covariates to edge-conditioned graph classification

EDGE DATA

1 Define context edges

Only observed training-graph links carry e_{ij}
Target candidate edge is absent
Keep resource → consumer orientation

2 Phase 1 edge vector

$r_{ij} = \log_{10}(m_consumer / m_resource)$
Mass-missing indicator
Optional explicit direction channel

3 Phase 2 enrichment

One-hot interaction type
Dimensionality and classification
Use only fields available at inference time

4 Train-only preprocessing

Fit imputation and scaling on training webs
Transform validation/test without refitting
Record provenance and missingness

edge schema frozen

SUBGRAPH ASSEMBLY

5 Reuse current candidates

Same positive splits, role-aware negatives
Same seeds and 1:1 class balance
No protocol change during ablation

6 Mask the target interaction

Remove source → target from adjacency
Remove its edge vector simultaneously
Never expose held-out interaction metadata

7 Extract directed h = 1 context

Preserve ordered endpoints and local directions
Attach e_{ij} to observed context edges
Retain 44 node attributes + node tags

8 Construct graph tensors

Node tensor: structural tag ⊕ attributes
Edge tensor: covariates + missingness
Directed edge index or paired direction channels

graph tensors

EDGE-CONDITIONED LEARNING

9 Edge network / gate

$F\theta(e_{ji})$ generates an edge-specific filter
or $g\theta(e_{ji})$ gates the neighbor message
Shared parameters across all subgraphs

10 Directional aggregation

$h_i^l = \sigma[\sum_j F\theta(e_{ji})\, h_j^{l-1} + W_o h_i^{l-1}]$
Separate incoming/outgoing transforms
Normalize by neighborhood size

11 Pool candidate subgraph

Concatenate message-passing layers
Use SortPooling or attention pooling
Keep ordered endpoint identity visible

12 Classify candidate link

MLP → P(resource → consumer)
Binary loss as in current SEAL_directed
Optionally calibrate probabilities post hoc

paired evaluation

EXPERIMENT & SAFEGUARDS

13 Frozen ablation ladder

A: structure only
B: + 44 node attributes
C: + mass-ratio edge channel
D: + richer interaction descriptors

14 Fair comparison

Reuse splits, seeds and negative samples
Match training budget and threshold policy
Report paired per-web differences

15 Leakage & missingness audit

Assert target-edge data are absent
Report feature coverage and fallback strata
Sensitivity analysis with missingness masks

16 Decision gate

Retain extension only if gains are robust
Separate existence from strength prediction
Document compute cost and uncertainty

IMPLEMENTATION ORDER 1) scalar edge gate with mass ratio → 2) true ECC-style vector filters → 3) richer descriptors only after coverage and leakage audits.

This design is a research proposal, not a description of the current repository. The present model remains binary-edge and weak-undirected inside DGCNN.

Method basis: Dynamic Edge-Conditioned Filters (Simonovsky & Komodakis, CVPR 2017)